CS251 - Homework 3

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Problem 2.4.2. Consider the sentence $\phi \overset{\text{def}}{=\!=} \forall x \exists y \exists z (P(x,y) \wedge P(z,y) \wedge (P(x,z) \to P(z,x)))$. Which of the following models satisfies ϕ?

a. The model $\mathcal{M}$ consists of the set of natural numbers with $P^{\mathcal{M}} \overset{\text{def}}{=\!=} \{(m,n) \mid m < n\}$.

b. The model $\mathcal{M}'$ consists of the set of natural numbers with $P^{\mathcal{M}'} \overset{\text{def}}{=\!=} \{(m, 2 * m) \mid m \text{ natural number}\}$.

c. The model $\mathcal{M}''$ consists of the set of natural numbers with $P^{\mathcal{M}''} \overset{\text{def}}{=\!=} \{(m,n) \mid m < n+1\}$

a. Suppose an arbitrary x. Choose $y = x + 1$; then it is readily apparent that $P(x,y)$. Choose $z = y - 1$; it is also obvious that $P(z,y)$. Observe that since $z = y - 1 = x$, we obtain $P(x,z)$ is false and thus $P(x,z) \to P(z,x)$ is true. Therefore, we yield that the model $\mathcal{M}$ satisfies ϕ.

b. Suppose an arbitrary x. Choose $y = 2x$ in order to satisfy $P(x,y)$, then $z = x$. Because $P(x,z)$ is false, the implication evaluates to true and thus the model $\mathcal{M}'$ satisfies ϕ.

c. Suppose an arbitrary x. Choose $y = x$ to satisfy $P(x,y)$ and choose $z = y$ to satisfy $P(z,y)$. Because $P(x,z)$ and $P(z,x)$ both evaluate to true, the implication does so as well and thus the model $\mathcal{M}''$ satisfies ϕ.

Problem 2.4.3. Let P be a predicate with two arguments. Find a model which satisfies the sentence $\forall x \neg P(x,x)$; also find one which doesn't.

Define $\mathcal{M}$ to be the model consisting of the set of natural numbers with $P^{\mathcal{M}} \overset{\text{def}}{=\!=} \{(m,n) \mid m < n\}$. This model satisfies the given formula, as $x < x$ is false for any arbitrary x. Additionally, define $\mathcal{M}'$ to be the model consisting of the set of natural numbers with $P^{\mathcal{M}'} \overset{\text{def}}{=\!=} \{(m,n) \mid m = n\}$. Because $x = x$ is true for any arbitrary x, this model does not safisfy the formula.

Problem 2.3.7. Show the semantic entailment $\forall x \neg \phi \vDash \neg \exists x \phi$; for that you have to take any model which satisfies $\forall x \neg \phi$ and you have to reason why this model must also satisfy $\neg \exists x \phi$. You should do this in a similar way to the examples in Section 2.4.2.

Proof. Suppose a model $\mathcal{M}$ satisfying $\forall x \neg \phi$; we obtain $\mathcal{M} \vDash_{[x \mapsto a]} \neg \phi$ holds for all a and thus $\mathcal{M} \vDash_{[x \mapsto a]} \phi$ holds for none of the a. This shows that $\mathcal{M} \nvDash_{[x \mapsto a]} \exists x \phi$ and so we arrive to $\mathcal{M} \vDash_{[x \mapsto a]} \neg \exists x \phi$.

Problem 2.4.8. Show the semantic entailment $\forall x P(x) \vee \forall x Q(x) \vDash \forall x (P(x) \vee Q(x))$.

Proof. Suppose a model $\mathcal{M}$ satisfying $\forall x P(x) \vee \forall x Q(x)$. Suppose $\mathcal{M} \vDash_l \forall x P(x)$, then $\mathcal{M} \vDash_{[x \mapsto a]} P(x)$ for all a. Thus one also can conclude $\mathcal{M} \vDash_{[x \mapsto a]} (P(x) \vee Q(x))$. By symmetry, this also works for the case $\mathcal{M} \vDash_l \forall x Q(x)$ and thus $\mathcal{M} \vDash_{[x \mapsto a]} (P(x) \vee Q(x))$. Consequently, $\mathcal{M} \vDash_l \forall x (P(x) \vee Q(x))$, as desired.

Problem 2.4.10. Is $\forall x(P(x) \vee Q(x)) \vDash \forall x P(x) \vee \forall x Q(x)$ a semantic entailment? Justify your answer.

The given string is not a semantic entailment. Let $\mathcal{M}$ be a model consisting of the set of natural numbers, where $P^{\mathcal{M}} = \{(m,n) \mid m < 0\}$ and $Q^{\mathcal{M}} = \{(m,n) \mid m \geq 0\}$. Clearly $\mathcal{M} \vDash_l \forall x(P(x) \vee Q(x))$. However, not every element is below 0 and not every element falls below 0 and so $\mathcal{M} \vDash \forall x P(x) \vee \forall x Q(x)$ does not hold.

Lemma 1.

$$\phi \to \psi \dashv\vdash \neg\phi \vee \psi$$

Proof. Well-known.

Lemma 2.

$$\neg(\phi \to \psi) \dashv\vdash \phi \wedge \neg\psi$$

Proof. Well-known.

Problem 2.4.12. For each of the formula of predicate logic below, either find a model which does not satisfy it, or prove it is valid:

a. $(\forall x \forall y(S(x,y) \to S(y,x))) \to (\forall x \neg S(x,x))$.

c. $(\forall x(P(x) \to \exists y Q(y))) \to (\forall x \exists y(P(x) \to Q(y)))$.

d. $(\forall x \exists y(P(x) \to Q(y))) \to (\forall x(P(x) \to \exists y Q(y)))$.

f. $(\forall x \forall y(S(x,y) \to (x = y))) \to (\forall x \neg S(z,z))$.

h. $\forall x \forall y((P(x) \to P(y)) \wedge (P(y) \to P(x)))$.

i. $(\forall x((P(x) \to Q(x)) \wedge (Q(x) \to P(x)))) \to ((\forall x P(x)) \to (\forall x Q(x)))$.

For every formula, we apply Gödel's completeness theorem to prove semantic validity via syntactic validity.

a. For ease of reference, let ϕ be the formula

$$(\forall x \forall y(S(x,y) \to S(y,x))) \to (\forall x \neg S(x,x)).$$

Then ϕ is not valid. Define by $\mathcal{M}$ the model consisting of natural numbers in which the predicate $S^{\mathcal{M}}$ is the equality predicate $=$. The antecedant is clearly true due to the commutativity of equality, yet the consequent is false due to the reflexivity of equality. Thus, we obtain that $\mathcal{M} \vDash_l \phi$ does not hold.

c. For ease of reference, let ϕ be the formula

$$(\forall x(P(x) \to \exists y Q(y))) \to (\forall x \exists y(P(x) \to Q(y))).$$

Then ϕ is valid. A proof follows.

1.	$\neg(\forall x(P(x) \to \exists y Q(y))) \to (\forall x \exists y(P(x) \to Q(y)))$	assumption
2.	$(\forall x(P(x) \to \exists y Q(y))) \wedge \neg\forall x \exists y(P(x) \to Q(y))$	(Lemma 2) 1
3.	$\forall x(P(x) \to \exists y Q(y))$	$\wedge e_1$ 2
4.	$\neg\forall x \exists y(P(x) \to Q(y))$	$\wedge e_2$ 2
5.	$\exists x(\neg\exists y(P(x) \to Q(y)))$	$\neg\forall$e 4
6.	x_0	assumption
7.	$\neg\exists y(P(x_0) \to Q(y))$	assumption
8.	$\forall y(\neg(P(x_0) \to Q(y)))$	$\neg\exists$e 7
9.	$P(x_0) \to \exists y Q(y)$	$\forall$e 3
10.	$\exists y(P(x_0) \to Q(y))$	(2.3.9b) 9
11.	y_0	assumption
12.	$P(x_0) \to Q(y_0)$	assumption
13.	$\neg(P(x_0) \to Q(y_0))$	$\forall$e 8
14.	$\perp$	$\neg$i 12, 13
15.	$\perp$	$\exists$e 10, 11–14
16.	$\perp$	$\exists$e 5, 6–15
17.	$(\forall x(P(x) \to \exists y Q(y))) \to (\forall x \exists y(P(x) \to Q(y)))$	PBC 1–16

d. For ease of referencee, let ϕ be the formula

$$(\forall x \exists y(P(x) \to Q(y))) \to (\forall x(P(x) \to \exists y Q(y))).$$

Then ϕ is valid. A proof follows.

1.	$\neg((\forall x \exists y(P(x) \to Q(y))) \to (\forall x(P(x) \to \exists y Q(y))))$	assumption
2.	$(\forall x \exists y(P(x) \to Q(y))) \wedge \neg\forall x(P(x) \to \exists y Q(y))$	(Lemma 2), 1
3.	$\forall x \exists y(P(x) \to Q(y))$	$\wedge e_1$ 2
4.	$\neg\forall x(P(x) \to \exists y Q(y))$	$\wedge e_2$ 2
5.	$\exists x(\neg(P(x) \to \exists y Q(y)))$	$\neg\forall$e 4
6.	x_0	assumption
7.	$\neg(P(x_0) \to \exists y Q(y))$	assumption
8.	$P(x_0) \wedge \neg\exists y Q(y)$	(Lemma 2), 1
9.	$P(x_0)$	$\wedge e_1$ 8
10.	$\neg\exists y Q(y)$	$\wedge e_2$ 8
11.	$\forall y \neg Q(y)$	$\neg\exists$e 10
12.	$\exists y(P(x_0) \to Q(y))$	$\forall$e 3
13.	y_0	assumption
14.	$P(x_0) \to Q(y_0)$	assumption
15.	$Q(y_0)$	$\to$e 14, 9
16.	$\neg Q(y_0)$	$\forall$e 11
17.	$\perp$	$\neg$i 15, 16
18.	$\perp$	$\exists$e 12, 13–17
19.	$\perp$	$\exists$e 5, 6–18
20.	$(\forall x \exists y(P(x) \to Q(y))) \to (\forall x(P(x) \to \exists y Q(y)))$	PBC 1–19

f. For ease of reference, let ϕ be the formula

$$(\forall x \forall y(S(x,y) \to (x = y))) \to (\forall x \neg S(z,z)).$$

Then ϕ is not valid. Define by $\mathcal{M}$ the model consisting of natural numbers in which the predicate $S^{\mathcal{M}}$ is the equality predicate $=$. The antecedant is obviously true, yet the consequent is false due to the reflexitivity of equality. Thus, we obtain that $\mathcal{M} \vDash_l \phi$ does not hold.

h. For ease of reference, let ϕ be the formula

$$\forall x \forall y((P(x) \to P(y)) \land (P(y) \to P(x))).$$

Then ϕ is not valid. Define by $\mathcal{M}$ the model consisting of natural numbers in which the predicate $P^{\mathcal{M}}$ is the greater-than predicate $>$. Choose the environment $l[x \mapsto 67, y \mapsto 69]$; then it's clear only one of the conjunction formula holds and thus the whole conjunction does not hold. We therefore obtain that $\mathcal{M} \vDash_{[x \mapsto 67, y \mapsto 69]} \phi$ does not hold.

i. For ease of reference, let ϕ be the formula

$$(\forall x((P(x) \to Q(x)) \land (Q(x) \to P(x)))) \to ((\forall x P(x)) \to (\forall x Q(x)))$$

Then ϕ is valid. A proof follows.

1.	$\neg((\forall x((P(x) \to Q(x)) \land (Q(x) \to P(x)))) \to ((\forall x P(x)) \to (\forall x Q(x))))$	assumption
2.	$(\forall x((P(x) \to Q(x)) \land (Q(x) \to P(x)))) \land \neg((\forall x P(x)) \to (\forall x Q(x)))$	(Lemma 2) 1
3.	$\forall x((P(x) \to Q(x)) \land (Q(x) \to P(x)))$	$\land e_1$ 2
4.	$\neg((\forall x P(x)) \to (\forall x Q(x)))$	$\land e_2$ 2
5.	$(\forall x P(x)) \land \neg(\forall x Q(x))$	(Lemma 2) 4
6.	$\forall x P(x)$	$\land e_1$ 5
7.	$\neg \forall x Q(x)$	$\land e_2$ 5
8.	$\exists x \neg Q(x)$	$\neg\forall$e 7
9.	x_0	assumption
10.	$\neg Q(x_0)$	assumption
11.	$(P(x_0) \to Q(x_0)) \land (Q(x_0) \to P(x_0))$	$\forall$e 3
12.	$P(x_0) \to Q(x_0)$	$\land e_1$ 11
13.	$\neg P(x_0)$	MT 12, 10
14.	$P(x_0)$	$\forall$e 6
15.	$\bot$	$\neg$i 14, 13
16.	$\bot$	$\exists$e 8, 9–15
17.	$(\forall x((P(x) \to Q(x)) \land (Q(x) \to P(x)))) \to ((\forall x P(x)) \to (\forall x Q(x)))$	PBC 1–16